

Malware Traffic Analysis Investigation Report

1. Objective

The objective of this investigation was to analyze suspicious network traffic using Wireshark and identify indicators of potentially malicious activity within a packet capture file.

2. Tools Used

- Wireshark
- Protocol Hierarchy Statistics - TCP Stream Analysis
- HTTP Object Export
- Packet Inspection Filters

3. Initial Traffic Observations

- Approximately 3292 packets were present in the capture.
- TCP was identified as the dominant protocol.
- HTTP traffic was observed throughout the capture.
- Minimal DNS activity was detected.
- Multiple communications with external IP addresses were identified.

4. Protocol Analysis

Protocol hierarchy analysis revealed significant TCP-based communication with noticeable HTTP traffic. The traffic pattern suggested active web communication and possible payload delivery activity.

Observed protocols included:

- HTTP
- DNS
- TCP
- IPv4

5. DNS Analysis

DNS filtering produced minimal results during the investigation. The absence of significant DNS traffic suggested the possibility of direct IP communication instead of domain-based communication.

This behavior can sometimes be associated with suspicious or malicious traffic attempting to avoid traditional DNS-based detection methods.

6. HTTP Traffic Analysis

HTTP analysis revealed suspicious web requests involving executable file downloads.

A suspicious GET request was identified:

- GET /jsaxo8u/g39b2cx.exe HTTP/1.1

Key observations:

- Executable file requested over HTTP
- Randomized file naming pattern observed
- Direct IP communication detected
- Unencrypted payload transfer identified

The communication appeared abnormal compared to standard browsing activity.

7. Suspicious Indicators Identified

Suspicious External IP: 185.91.175.64

Suspicious File: g39b2cx.exe

Suspicious Activity:

- HTTP-based executable download
- Binary payload transfer
- Direct IP communication
- Repeated communication patterns

Content-Type Observed: application/octet-stream

8. TCP Stream Analysis

- TCP stream reconstruction was performed on the suspicious HTTP communication.
- The reconstructed stream revealed executable payload content and included the following indicator:
- "This program cannot be run in DOS mode"
- This strongly indicated the transfer of a Windows executable file through the observed HTTP session.

9. Exported HTTP Objects

- HTTP object analysis identified transferred files associated with the suspicious communication.
- The exported objects further supported the observation of executable payload delivery through HTTP traffic. No files were executed during the investigation process.

10. Packet Inspection Findings

- Deep packet inspection confirmed:
- Source and destination IP communication
- HTTP communication over port 80
- Executable payload delivery behavior
- Suspicious GET request patterns

The packet details aligned with previously identified suspicious indicators.

11. Indicators of Compromise (IOCs)

Type	Value	Observation
External IP	185.91.175.64	Suspicious communication host
Executable File	g39b2cx.exe	Potential malicious payload
Protocol HTTP	Unencrypted communication	
Content-Type	application/octet-stream	Binary executable transfer
HTTP Request	GET /jsaxo8u/g39b2cx.exe	Suspicious payload request

12. Conclusion

The investigation revealed multiple suspicious indicators consistent with potential malware delivery activity.

Analysis of the packet capture identified:

- Suspicious HTTP communication
- Executable payload transfer
- Binary file delivery
- Direct IP communication
- Suspicious external host interaction

The observed behavior strongly suggested potentially malicious activity involving executable delivery over HTTP traffic.

This investigation demonstrated the use of Wireshark for packet analysis, protocol investigation, TCP stream reconstruction, IOC identification, and malware traffic analysis workflows.